

Education & Skills

B.S. Computer Science and Engineering, GPA: 3.94. University of Louisville.

Aug. 2020 – May 2024

Skills: Product Management, AI, Machine Learning, Cloud Infrastructure and Security (Azure), Data Pipelines, Cross-Functional Collaboration, Model Evaluation, Product Metric Analysis, Recommender Systems, Python, C#, TypeScript, Power Automate.

Work Experience

Software Engineer (SWE)

Microsoft

Jul. 2024 – Present

- Built Azure infrastructure for dataset management, AI workflows, and speech model training across multiple teams.
- Designed KPI dashboards and built data pipelines to migrate 18PB of data from legacy storage to modern infrastructure.
- Implemented a custom RBAC security framework, expanding adoption across teams and enhancing compliance.
- Developing a Copilot agent to assess and profile training data, aligning team efforts via OKRs.

Project Manager and Researcher

University of Hawaii at Hilo

Jan. 2023 – Dec. 2023

- Secured \$6k grant through a proposal outlining project scope, timeline, and budget for AI-driven conservation initiative.
- Evaluated deep learning models on classifying endangered birds, assessing the impact of class variation on performance.
- Assessed model limitations, highlighting misclassification risks from dataset bias, and proposed mitigation strategies.
- Presented findings to stakeholders, highlighting AI monitoring's disruption of manual ecological survey methods.

SWE Intern

Microsoft

Aug. 2023 – Nov. 2023

- Enhanced OneDrive's AI image search by developing a PyTorch-based tool to scale datasets, improving search relevance.
- Improved dataset quality with metadata encoding through location-tagged images and location-based queries.
- Analyzed geographic distribution of OneDrive users and generated test set reflecting user demographics.

Product Manager Intern

University of Louisville

May 2023 – Aug. 2023

- Led cross-team efforts to define user requirements and design Figma wireframes for a student performance analysis tool.
- Researched FERPA compliance and scoped cloud architecture, maintaining privacy regulations and budget constraints.
- Engaged directly with customers to present design progression and make iterative prototype improvements.

Product Manager and SWE Intern

Microsoft

May 2022 – Aug. 2022

- Led ideation, design, and implementation of Microsoft Teams extension aimed at connecting remote coworkers.
- Defined user personas and stakeholder requirements to guide application and wireframe design in Figma.
- Developed a React/TypeScript demo integrated with Azure Cloud Storage for scalable user data management.

Applied Experience

Research Assistant

University of Louisville

Jan. 2021 – Aug. 2023

- Mitigated popularity bias in collaborative filtering recommender systems (RSs), enhancing content discovery for users.
- Introduced novel popularity metric and developed a multi-armed bandit-based selection mechanism.
- Analyzed 10 RSs across 7 fairness and performance metrics to address system biases and drive novel user experiences.

Computer Science Coloring Book Author

Jan. 2022 – Jan. 2023

- Authored and illustrated educational computer science coloring book introducing software concepts to young students.
- Secured \$4.8k grant to distribute 1.9k copies to rural communities impacted by poverty.
- Launched supporting website to expand accessibility with free digital resources: <https://www.coloringcomputing.org/>